DATA STRUCTURES — I

Information needs organization, like objects in real life.

SCAN & LEARN

WHAT HAPPENS • FOLLOW THE ARROWS

01 YOU HAVE VALUES → 02 ORGANIZE THEM → 03 CHOOSE A RULE → 04 FIND THEM EASILY

FOUR IDEAS TO KEEP

1 LIST = ITEMS IN ORDER

Like seats on a train: each item has a position. In Python the first position is 0, not 1. You can visit or change items.

["Ana", "Dan", "Mia"] → position 0 is “Ana”.

2 STACK = STACK OF PLATES

Add on top, remove from the top. Last In, First Out: LIFO. Useful for Undo and function calls.

Push A, then B, then C → pop C first.

3 QUEUE = CHECKOUT LINE

First In, First Out: FIFO. Items join at the back and leave from the front. Useful for waiting tasks.

A, B, C join → A leaves first.

4 SET = UNIQUE ITEMS

Stores values without duplicates. Their order is not guaranteed. Useful for “Have I seen this name before?”.

[A, B, A] in a set → {A, B}.

TRY IT — A CLASS WITHOUT SCREENS

Three students are A, B, C. Use where you stand to show removal order for a stack and queue.

THE ONE SENTENCE TO REMEMBER

LIST = position · STACK = last · QUEUE = first · SET = no duplicates.

Question: Which structure resembles the Undo button?